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95 - Numerical analysis of deformations in deep twin tunnels with a transverse gallery

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Abstract Submission

Resumo

The increasing construction of twin tunnels necessitates a thorough understanding of their mechanical interactions with transverse galleries and surrounding rock masses. This study presents a 3D finite element analysis of deep twin tunnels connected by a transverse gallery, incorporating time-dependent effects such as concrete creep, shrinkage, and rock mass creep. The computational model is verified against prior numerical studies, demonstrating its reliability in predicting lining internal forces. Results reveal significant anisotropy in lining pressures and wall deformations, influenced by lining type (elastic vs. viscoelastic) and rock mass friction angle. Maximum deformations occur at the tunnel crown, indicating horizontal ovalization, while minimum deformations - and corresponding peak pressures - appear near the tunnel sidewall, particularly at $\theta = 0^\circ$. Higher friction angles reduce convergence and pressures at this location, mainly at the end of construction. Viscoelastic lining exhibit greater long-term redistribution of both deformations and pressures. The study highlights the critical role of 3D modeling and time-dependent material behavior in optimizing tunnel design for long-term stability, providing practical insights for safer and more efficient underground infrastructure.

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